# Composite functions

**1**

**a** $\dfrac{dy}{dx} = e^{x}\left(\sin x + \cos x\right)$

**b** $\dfrac{dy}{dx} = e^{\sin x}\cos x$

**c** $\dfrac{dy}{dx} = e^{x}\left(\cos 3x - 3\sin 3x\right)$

**d** $\dfrac{dy}{dx} = e^{3x}\left(3\cos\left(\dfrac{x}{3}\right) - \dfrac{1}{3}\sin\left(\dfrac{x}{3}\right)\right)$

**e** $\dfrac{dy}{dx} = e^{-x}\left(4\cos 4x - \sin 4x\right)$

**f** $\dfrac{dy}{dx} = \dfrac{-\sin\left(\ln x\right)}{x}$

**g** $\dfrac{dy}{dx} = \dfrac{-\sin\left(\ln\left(-5x\right)\right)}{x}$

**h** $\dfrac{dy}{dx} = -\sin x \ln x + \dfrac{\cos x}{x}$

**i** $\dfrac{dy}{dx} = e^{x}\left(\ln x + \dfrac{1}{x}\right)$

**j** $\dfrac{dy}{dx} = e^{2x}\left(2\ln\left(4x\right) + \dfrac{1}{x}\right)$

**k** $\dfrac{dy}{dx} = \dfrac{1}{x\ln x}$

**l** $\dfrac{dy}{dx} = \dfrac{1 - 2x\ln x}{xe^{2x}}$

**m** $\dfrac{dy}{dx} = -\dfrac{3}{e^{2}x(1 - 6x^{3})}$

**n** $\dfrac{dy}{dt} = -\dfrac{\pi}{2}\sin\left(\dfrac{\pi t}{2} - \dfrac{\pi}{3}\right)$

**o** $\dfrac{dy}{dx} = -\cos\left(\dfrac{1}{x}\right) + 2x\sin\left(\dfrac{1}{x}\right)$

**p** $\dfrac{dy}{dx} = \dfrac{4}{5}\left(\cos\left(\dfrac{x}{5}\right) - 15e^{2x} - 10x^{-9}\right)$

**q** $\dfrac{dy}{dx} = \dfrac{2x - 7}{x^{2} - 7x - 12} - \dfrac{\sqrt{11}}{2\sqrt{x}}$

**r** $\dfrac{dy}{dx} = -5\left(e^{-5x^{2}} + \cos x\right)^{4}\left(10xe^{-5x^{2}} + \sin x\right)$

**s** $\dfrac{dy}{dx} = \dfrac{e^{-0.2x}\left(0.2x^{4} + 4x^{3} - 0.2\sin\left(\dfrac{\pi}{4}x\right) - \dfrac{\pi}{4}\cos\left(\dfrac{\pi}{4}x\right)\right)}{\left(\sin\left(\dfrac{\pi}{4}x\right) - x^{4}\right)^{2}}$

**t** $\dfrac{dy}{dx} = e^{6x}\left(7\cos x + 5\sin x\right)$

**u** $\dfrac{dy}{dx} = \dfrac{-4\sin\left(4x - \dfrac{10\pi}{11}\right)(x+3) - 2\cos\left(4x - \dfrac{10\pi}{11}\right)}{(x+3)^{3}}$

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**2**

**a** $e^{5x}\left(5\cos x - \sin x\right)$

**b** $x^3\cos x + 3x^2\sin x$

**c** $-\sin 2x$

**d** $\dfrac{-2}{1+\sin 2x}$

**e** $\dfrac{4\sin x - 4x\cos x}{\sin^2(x)}$

**f** $e^{3x}\left(3\cos\left(5x+\dfrac{4\pi}{7}\right) - 5\sin\left(5x+\dfrac{4\pi}{7}\right)\right)$

**g** $\dfrac{\sin x - x\ln x\cos x}{x\sin^2(x)}$

**3**

**a**

**i** $-3\sin 3x$ **ii** $-\dfrac{3}{2}$

**b**

**i** $3\sin 3x$ **ii** $\dfrac{3\sqrt{3}}{2}$

**c**

**i** $4\cos 4x$ **ii** $2\sqrt{2}$

**d**

**i** $4\sin 8x$ **ii** $2\sqrt{2}$

**e**

**i** $-2\sin 4x$ **ii** $-1$

**4**

**a** $\dfrac{dy}{dx} = -e^{\cos x}\sin x$

**b** $\dfrac{dy}{dx} = e^x\cos\left(e^x\right)$

**c** $\dfrac{dy}{dx} = e^{\cos x}\left(e^x\cos\left(e^x\right) - \sin x\sin\left(e^x\right)\right)$

**5**

**a** $u' = 8x + e^x$

**b** $v' = -7\sin\left(7x\right)$

**c** $\dfrac{d}{dx}\left(\dfrac{4x^2+e^x}{\cos 7x}\right) = \dfrac{8x\cos 7x + e^x\cos 7x + 28x^2\sin 7x + 7e^x\sin 7x}{\cos^2(7x)}$

**6**

**a** No solution provided.

**b** $\dfrac{dy}{dx} = (x+1)\,e^x$

**7** $\dfrac{dy}{dx} = \dfrac{5\left(\ln(x+5)\right)^4}{x+5}$

**8**

**a** $\dfrac{dy}{dx} = \dfrac{-2\left(1+\ln\left(0.3x\right)\right)^{-3}}{x}$

**b** $\dfrac{dy}{dx} = \dfrac{-2e^x\left(1+\ln\left(0.3x\right)\right)^{-3}}{x} + e^x\left(1+\ln\left(0.3x\right)\right)^{-2}$

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**9** a $\frac{d}{dx}\left(2\ln\left(-3x-x^2\right)\right)=\frac{2\left(-3-2x\right)}{-3x-x^2}$

b $\frac{dy}{dx}=3\left(2\ln\left(-3x-x^2\right)-\cos\left(\frac{\pi}{3}x\right)+\frac{1}{x^4}\right)^2\left(\frac{2\left(-3-2x\right)}{-3x-x^2}+\frac{\pi}{3}\sin\left(\frac{\pi}{3}x\right)-\frac{4}{x^5}\right)$

## Stationary points

**10** a $f'\left(x\right)=3e^{2x}\cos 3x+2e^{2x}\sin 3x$

b $x=0,\frac{\pi}{3}$

c $(0.72,3.51)$

**11** a $f'(x)=3x^2\ln x+x^2$

b $x=e^{-\frac{1}{3}}$

c

| $x$ | $e^{-\frac{1}{2}}$ | $e^{-\frac{1}{3}}$ | $e^{-\frac{1}{4}}$ |
|---|---|---|---|
| $f(x)$ | $-\frac{1}{2}e^{-1}$ | $0$ | $\frac{1}{4}e^{\frac{1}{2}}$ |

d Minimum at $x=e^{-\frac{1}{3}}$

e $-\frac{1}{3e}$

f $-\frac{1}{2e^{\frac{3}{2}}}$

**12** a $x=\frac{1}{3}$

b $\frac{dy}{dx}=\frac{1-3x\ln 3x}{xe^{3x}}$

c No solution provided.

d

| $x$ | 0.01 | 0.58 | 1.58 |
|---|---|---|---|
| $\frac{dy}{dx}$ | 107.3 | 0 | -0.04 |

e Maximum

f $(0.58,0.10)$

**13** $b=-4$ and $a=4e$

**14** a $0<x<\frac{\pi}{2},\pi<x<\frac{3\pi}{2}$

b $x=\frac{\pi}{4},\frac{5\pi}{4}$

c $f\left(\frac{\pi}{4}\right)=0$

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## Applications

**15** $y = -\frac{\pi}{2}x + \frac{\pi^2}{4}$

**16**

**a** $y = e^{\frac{3\pi}{2}}x - \frac{3\pi e^{\frac{3\pi}{2}}}{2} + e^{\frac{3\pi}{2}} + 3$

**b** $y = x + 1 - \frac{3\pi}{2}$

**17**

**a** $P(0) = 4420$

**b** $P'(0) = 291$

**c** $P'\left(\frac{7}{12}\right) = 318$

**d** $P'(10) = 1036$

**18**

**a** $x(0) = 29 \text{ m}$

**b** $v(t) = (2t + \cos(t - 4\pi) - 1) \text{ m/s}$

**c** $a(t) = (2 - \sin(t - 4\pi)) \text{ m/s}^2$

**19**

**a** $x = 0$

**b** $\frac{dx}{dt} = 2\pi e^{2t}\cos\left(\pi e^{2t}\right)$

**c** $t = \frac{1}{2}\ln\frac{3}{2}, \frac{1}{2}\ln\frac{5}{2}$

**d** $x = -1$ means 1 unit left of the origin.

**e** $x = 1$ means 1 unit right of the origin.

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